

Database Principles

CSC3400

Chapter 4:

Database Normalization

Chapter Outline

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce Codd Normal Form (BCNF)
- Multivalued Dependency and Fourth Normal Form (4NF)

Data Normalization

- Primarily a tool to validate and improve a logical design so that it satisfies certain constraints that ***avoid unnecessary duplication of data***
- The process of decomposing relations with anomalies to produce smaller, ***well-structured*** relations

Recap...

Definition: A relation is a named, two-dimensional table of data

- Table is made up of rows (records), and columns (attribute or field)

Not all tables qualify as relations

Requirements:

- Every relation has a unique name.
- Every attribute value is atomic (not multivalued, not composite)
- Every row is unique (can't have two rows with exactly the same values for all their fields)
- Attributes (columns) in tables have unique names

Well-Structured Relations

- A relation that contains minimal data redundancy and allows users to insert, delete, and update rows without causing data inconsistencies
- Goal is to avoid anomalies
 - ▣ **Insertion Anomaly** – adding new rows forces user to include the details of that rows.
 - ▣ **Deletion Anomaly** – deleting rows may cause a loss of data that would be needed for other future rows
 - ▣ **Modification Anomaly** – changing data in a row forces changes to other rows because of duplication

Figure 4.1 Employee table

EMPLOYEE2					
<u>Emp_ID</u>	Name	Dept_Name	Salary	<u>Course_Title</u>	Date_Completed
100	Margaret Simpson	Marketing	48,000	SPSS	6/19/200X
100	Margaret Simpson	Marketing	48,000	Surveys	10/7/200X
140	Alan Beeton	Accounting	52,000	Tax Acc	12/8/200X
110	Chris Lucero	Info Systems	43,000	SPSS	1/12/200X
110	Chris Lucero	Info Systems	43,000	C++	4/22/200X
190	Lorenzo Davis	Finance	55,000	C++	7/23/200x
150	Susan Martin	Marketing	42,000	SPSS	6/19/200X
150	Susan Martin	Marketing	42,000	Java	8/12/200X

Question – Is this a relation?

Answer – Yes: unique rows and no multivalued attributes

Question – What's the primary key?

Answer – Composite: Emp_ID, Course_Title

Anomalies in this Table

Insertion – cannot enter a new employee without having the employee taking a course (since `course_title` is part of the primary key)

Deletion – if we remove employee 140, we lose information about the existence of a Tax Acc course

Modification – giving a salary increase to employees (eg. `Emp_id 100`) forces us to update multiple records

Why do these anomalies exist?

Because we've combined two themes (entity types) into one relation. This results in duplication, and an unnecessary dependency between the entities

Functional dependency

Functional dependency (FD) is a set of constraints between two attributes in a relation. Functional dependency says that if two tuples have same values for attributes $A_1, A_2, \dots, A_n$, then those two tuples must have to have same values for attributes $B_1, B_2, \dots, B_n$.

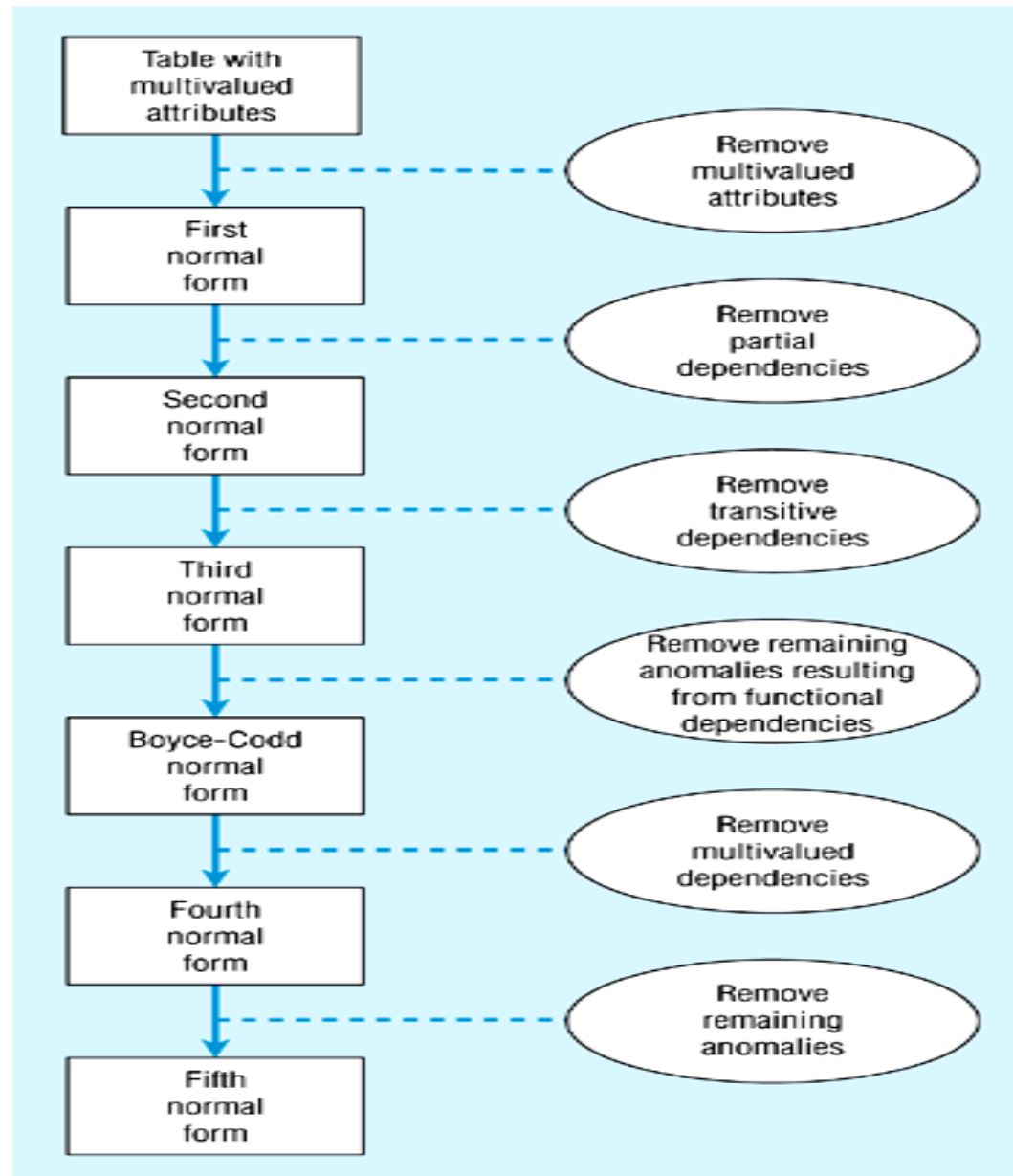
Functional dependency is represented by an arrow sign ($\rightarrow$) that is, $X \rightarrow Y$, where X functionally determines Y . The left-hand side attributes determine the values of attributes on the right-hand side.

Candidate Key:

- A unique identifier. One of the candidate keys will become the primary key
 - E.g. perhaps there is both credit card number and SS# in a table...in this case both are candidate keys
- Each non-key field is functionally dependent on every candidate key

<https://www.geeksforgeeks.org/introduction-of-database-normalization/>

Figure 4.2
Steps in
normalization



First Normal Form (1NF)

- Disallows

- composite attributes

- multivalued attributes

- nested relations; attributes whose values for an individual tuple are non-atomic



- Considered to be part of the definition of a relation

- Most RDBMSs allow only those relations to be defined that are in First Normal Form

First Normal Form (1NF)

(* not in 1NF-composite attribute)

EMPLOYEE_PROJECT

		Projects	
<u>SSN</u>	EmpName	ProjecNumber	Hours
123	Ahmad Ramli	1	32.5
668	Chan Sui Lin	3	40

❖ Remove composite attribute -Projects

EMPLOYEE_PROJECT (1NF)

<u>SSN</u>	EmpName	ProjecNumber	Hours
123	Ahmad Ramli	1	32.5
668	Chan Sui Lin	3	40

First Normal Form (1NF)

(* not in 1NF - multi-valued attribute)

DEPARTMENT

<u>DeptNumber</u>	Dname	DeptMgr_SSN	DeptLocation
10	Research	789	{Melaka, Sarawak, Terengganu}
20	Administration	562	Selangor
30	Headquarters	891	Kuala Lumpur

❖ Insert new tuples for each value of the attribute

DEPARTMENT (1NF)

<u>DeptNumber</u>	Dname	DeptMgr_SSN	<u>DeptLocation</u>
10	Research	789	Melaka
10	Research	789	Sarawak
10	Research	789	Terengganu
20	Administration	562	Selangor
30	Headquarters	891	Kuala Lumpur

First Normal Form (1NF)

(* not in 1NF – nested relations with repeating groups)

EMPLOYEE_PROJECT

<u>SSN</u>	EmpName	<u>ProjecNumber</u>	Hours
123	Ahmad Ramli	1	32.5
		2	7.5
668	Chan Sui Lin	3	40
456	Anita Ramasami	1	20
		2	20
335	Hanisah Bahrin	2	10
		3	10
		10	10

nested relations: EMPLOYEE_PROJECT(SSN, EmpName(ProjectNumber, Hours))

First Normal Form (1NF)

For the nested relation with repeating groups, expand the table to become a relation with atomic values

EMPLOYEE_PROJECT

<u>SSN</u>	EmpName	<u>ProjecNumber</u>	Hours
123	Ahmad Ramli	1	32.5
123	Ahmad Ramli	2	7.5
668	Chan Sui Lin	3	40
456	Anita Ramasami	1	20
456	Anita Ramasami	2	20
335	Hanisah Bahrin	2	10
335	Hanisah Bahrin	3	10
335	Hanisah Bahrin	10	10

A relation with atomic values, but with repeating groups, therefore can be split further
As EMPLOYEE(SSN, EmpName) and EMPLOYEE_PROJECT(SSN, ProjectNumber, Hours)

First Normal Form (1NF)

EMPLOYEE

<u>SSN</u>	EmpName
123	Ahmad Ramli
335	Hanisah Bahrin
456	Anita Ramasami
668	Chan Sui Lin

EMPLOYEE_PROJECT

<u>SSN</u>	<u>ProjecNumber</u>	Hours
123	1	32.5
123	2	7.5
668	3	40
456	1	20
456	2	20
335	2	10
335	3	10
335	10	10

What is atomic value?

This mean that in your table, for every row-by-column position (cell), there exists only one value - not an array or list of values:

OrderID	CustID	Date	Items
1	4	4/11/02	5 Pencils, 3 Erasers, 6 Rulers
2	23	6/11/02	1 Scissor
3	15	7/11/02	2 Pen, 2 Notebook
4	2	7/11/02	15 5" Magazine File
5	23	7/11/02	1 Stapler
6	2	8/11/02	5 Kingston USB Flash Drive 8GB

Attribute 'Items' is not atomic

Atomic Value: 1 cell 1 value

OrderID	CustID	Date	Quantity	Items
1	4	4/11/2002	5	Pencils
1	4	4/11/2002	3	Erasers
1	4	4/11/2002	6	Rulers
2	23	6/11/2002	1	Scissor
3	15	7/11/2002	2	Pen
3	15	7/11/2002	2	Notebook
4	2	7/11/2002	15	5" Magazine File
5	23	7/11/2002	1	Stapler
6	2	8/11/2002	5	Kingston USB Flash Drive 8GB

Example: Dept_Emp

DEPT_NO	MANAGER_NO	EMP_NO	EMP_NAME
D101	12345	20000 20001 20002	Carl Sagan Magic Johnson Larry Bird
D102	13456	30000 30001	Jimmy Carter Paul Simon

This table is NOT a RELATION, emp_no and emp_name are multi-valued.

...and now, is this a relation?

Dept_No	Manager_No	<u>Emp_No</u>	Emp_Name
D101	12345	2000	Carl Sagan
D101	12345	2001	Magic Johnson
D101	12345	2002	Larry Bird
D102	13456	3000	Jimmy Carter
D102	13456	3001	Paul Simon



- By filling in all values, this Dept_Emp table is now a relation since there are no multi-valued attribute ie. only atomic values in each cell.
- Assume the Primary key is composite Dept_No, Emp_No
- Repeating groups?

First Normal Form (1NF) - continue

If a relation has no repeating groups, it is said to be in first normal form (1NF)

However, this relation does contain repeating group:

DEPT_EMP(Dept No, Manager_No, (Emp_No, Emp_Name))

Therefore, split them by creating separate relations

- Everything that is outside the repeating group
 - ie. DEPT(Dept No, Manager_No)
- The repeating group attributes combine with the key attributes outside the group EMP(DeptNo, Emp_No, Emp_Name)

Second Normal Form (2NF)

- 1NF and no partial functional dependency
- Uses the concepts of FDs, primary key
- Definitions
 - **Prime attribute**: An attribute that is member of the primary key K
 - **Non-prime attribute** – An attribute, which is not a part of the prime-key, is said to be a non-prime attribute.
 - **Full functional dependency** : a FD $Y \rightarrow Z$ where removal of any attribute from Y means the FD does not hold anymore

Second Normal Form (2NF)

EMPLOYEE_PROJECT

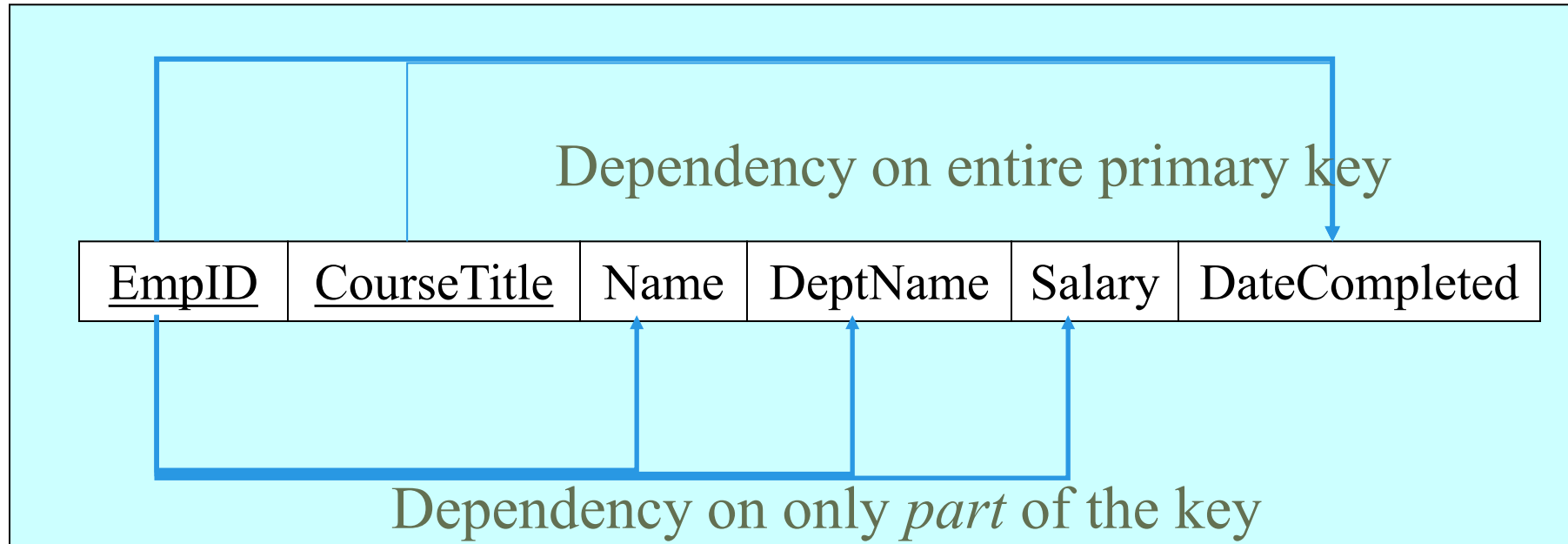
<u>SSN</u>	<u>ProjectNumber</u>	Hours
123	1	32.5
123	2	7.5
668	3	40
456	1	20
456	2	20
335	2	10
335	3	10
335	10	10

- The primary key (PK) is composite of SSN,ProjectNumber
- Hours which is a non-prime attribute (non-key) is FULLY functionally dependent on the PK
- Hours cannot be determined by only SSN
- Hours cannot be determined by only ProjectNumber
- Therefore there is no PARTIAL functional dependencies

Second Normal Form (2NF)

- A relation schema R is in second normal form(2NF) if every non-prime attribute A in R is fully functionally dependent on the primary key
 - R can be decomposed into 2NF relations via the process of 2NF normalization or “second normalization”
- Fig. 4.1 is NOT in 2nd Normal Form (see fig 4.3)

Figure 4.3— Functional Dependencies in EMPLOYEE2



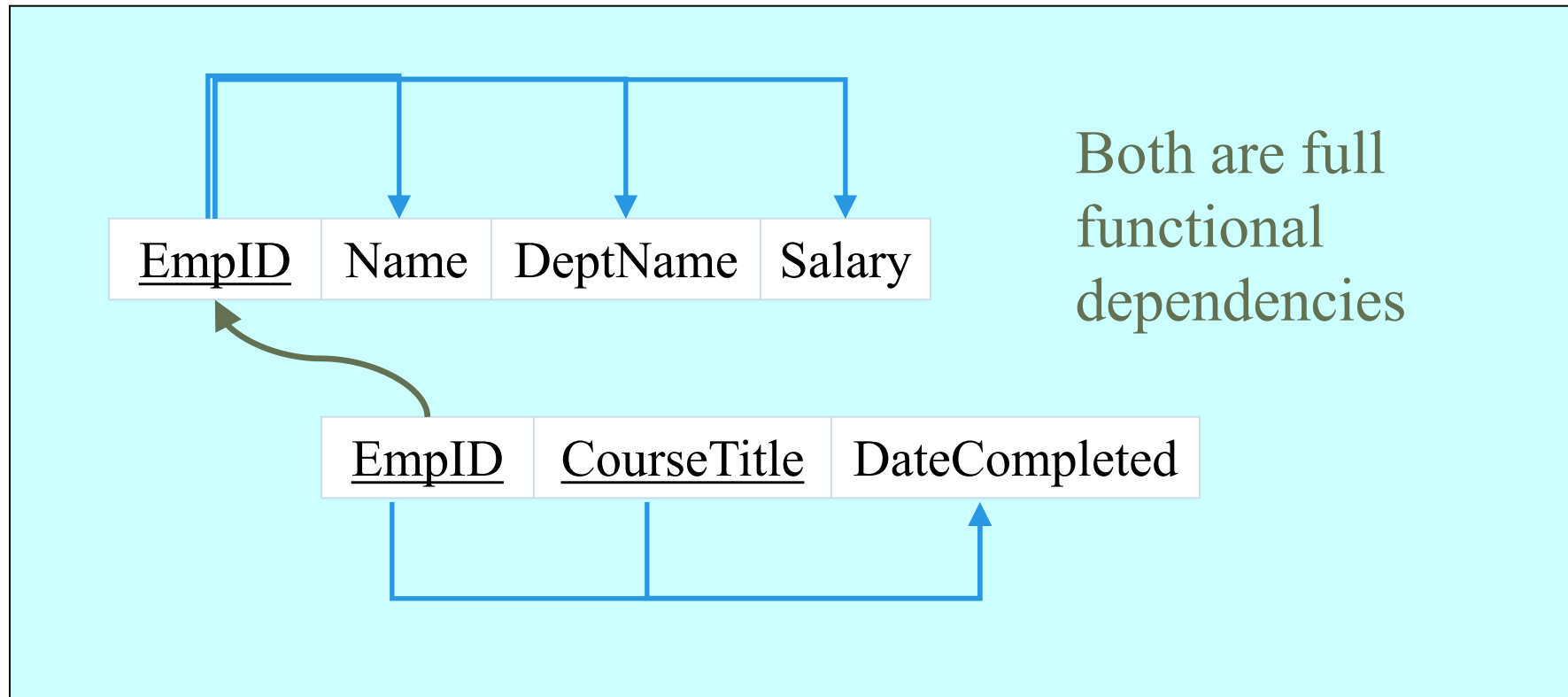
EmpID, CourseTitle → DateCompleted

EmpID → Name, DeptName, Salary

Therefore, NOT in 2nd Normal Form!!

Getting it into 2nd Normal Form

Figure 4.4 – decomposed into two separate relations



Third Normal Form

2NF PLUS *no transitive dependencies* (one attribute functionally determines a second, which functionally determines a third)

Definition:

Transitive functional dependency :a FD $X \rightarrow Z$ that can be derived from two FDs $X \rightarrow Y$ and $Y \rightarrow Z$

Third Normal Form (3NF)

- A relation schema R is in third normal form (3NF) if it is in 2NF and no non-prime attribute A in R is transitively dependent on the primary key
- R can be decomposed into 3NF relations via the process of 3NF normalization

NOTE:

- In $X \rightarrow Y$ and $Y \rightarrow Z$, with X as the primary key, we consider this a problem only if Y is not a candidate key.
- When Y is a candidate key, there is no problem with the transitive dependency

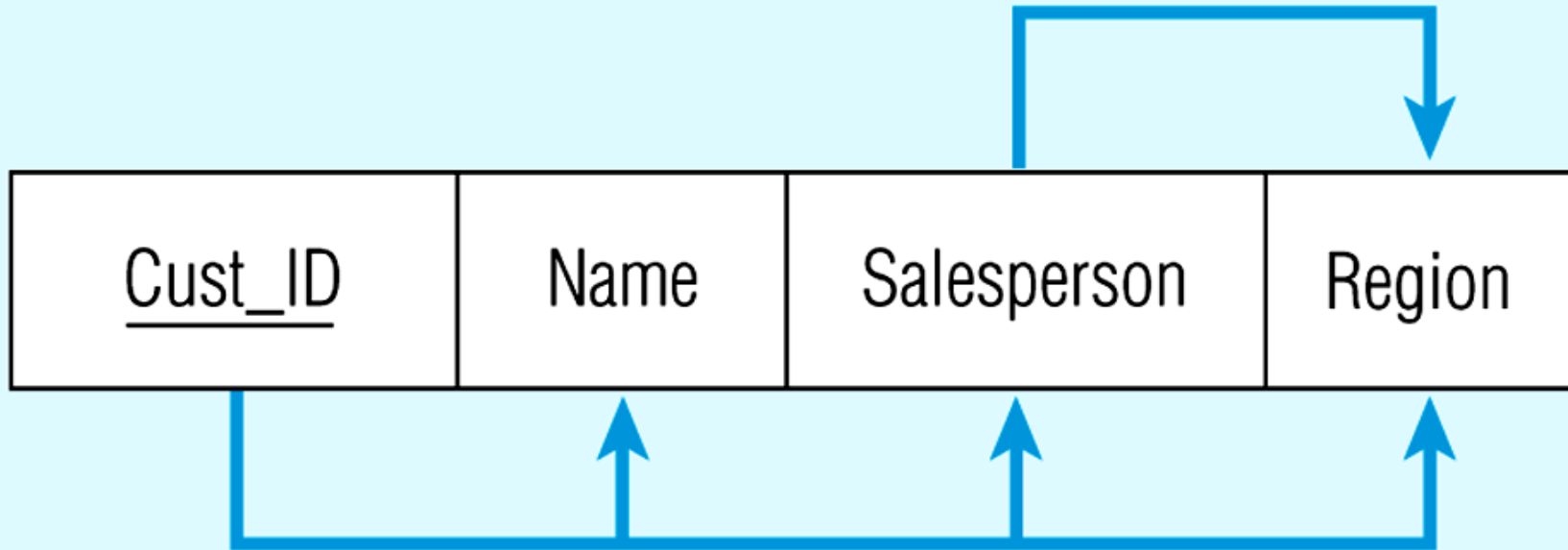
Fig. 4.5,4.6

Figure 4.5-- Relation with transitive dependency

(a) SALES relation with simple data

SALES			
Cust_ID	Name	Salesperson	Region
8023	Anderson	Smith	South
9167	Bancroft	Hicks	West
7924	Hobbs	Smith	South
6837	Tucker	Hernandez	East
8596	Eckersley	Hicks	West
7018	Arnold	Faulb	North

(b) Relation with transitive dependency



CustID $\rightarrow$ Name
CustID $\rightarrow$ Salesperson
CustID $\rightarrow$ Region

All this is OK
(2nd NF)

BUT, on assumption Salesperson is unique:

CustID $\rightarrow$ Salesperson $\rightarrow$ Region

Transitive dependency
(not 3rd NF)

Figure 4.6 Removing a transitive dependency

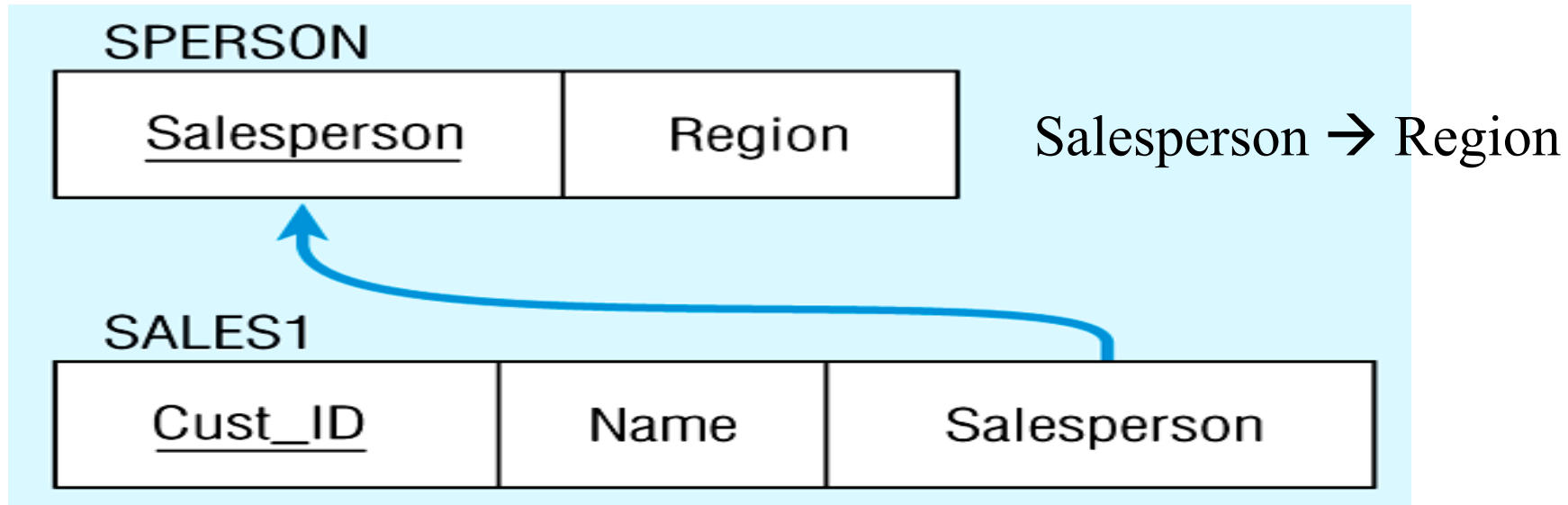
(a) Decomposing the SALES relation

SALES1			SPERSON	
Cust_ID	Name	Salesperson	Salesperson	Region
8023	Anderson	Smith	Smith	South
9167	Bancroft	Hicks	Hicks	West
7924	Hobbs	Smith	Hernandez	East
6837	Tucker	Hernandez	Faulb	North
8596	Eckersley	Hicks		
7018	Arnold	Faulb		

SALES1(Cust_ID, Name, Salesperson)

SPERSON(Salesperson, Region)

Figure 4.6(b) Relations in 3NF



CustID $\rightarrow$ Name

CustID $\rightarrow$ Salesperson

Now, there are no transitive dependencies...

Both relations are in 3rd NF

Normalization into 2NF and 3NF -
example from Ramez Elmasri and
Shamkant Navathe (2007)

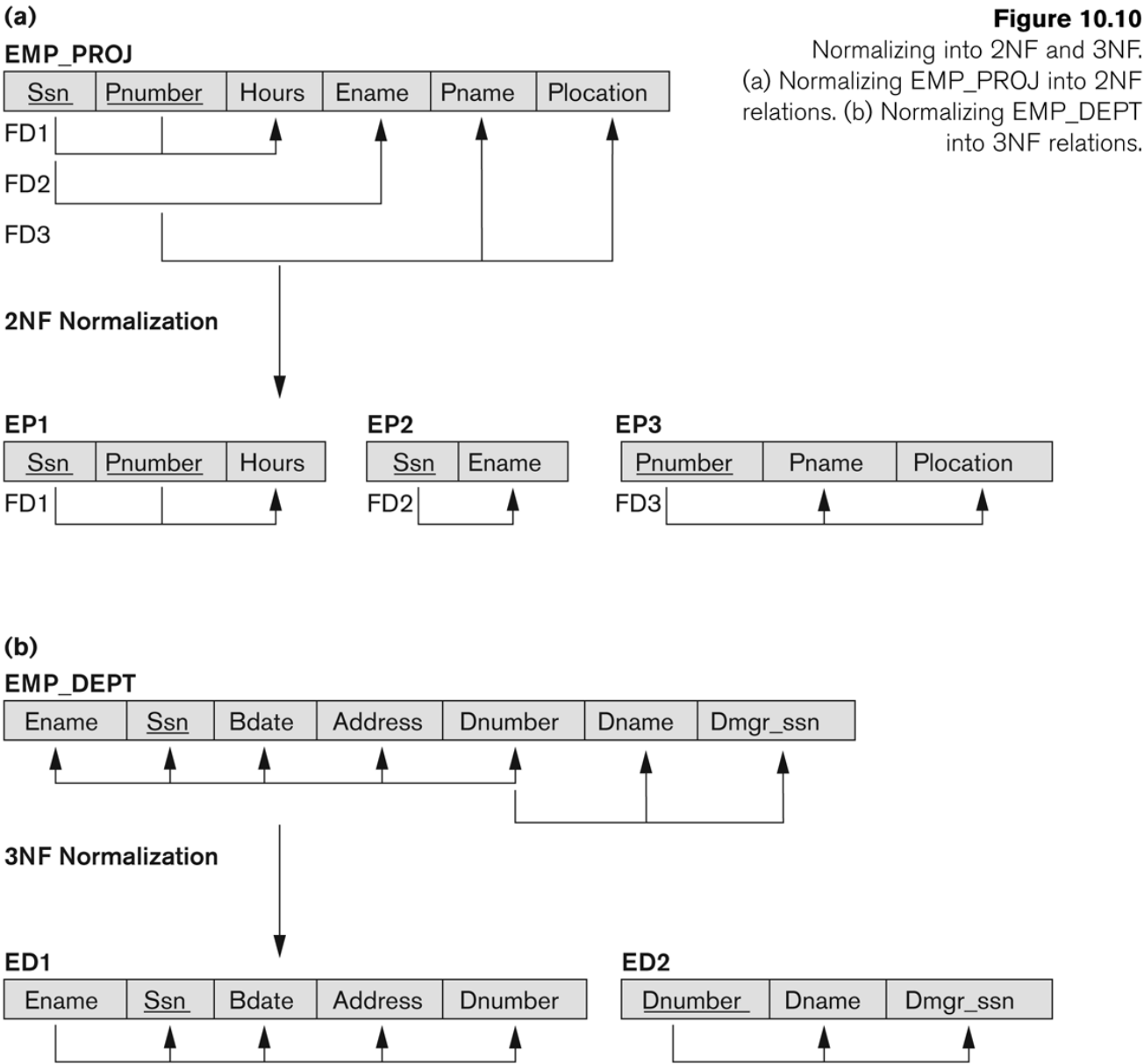


Figure 10.10
Normalizing into 2NF and 3NF.
(a) Normalizing EMP_PROJ into 2NF
relations. (b) Normalizing EMP_DEPT
into 3NF relations.

Normalization into 2NF and 3NF - example from Ramez Elmasri and Shamkant Navathe (2007)

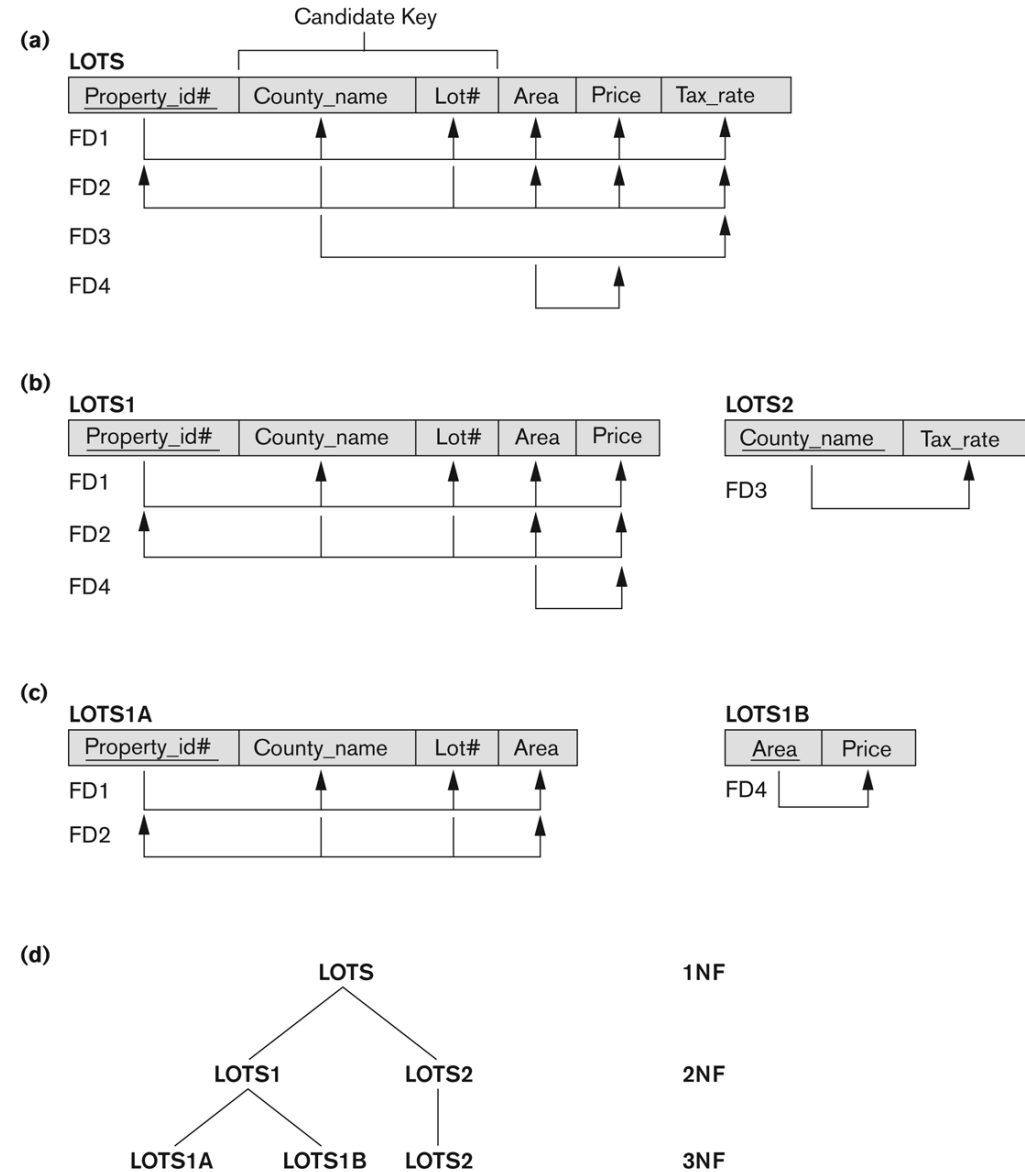


Figure 10.11

Normalization into 2NF and 3NF. (a) The LOTS relation with its functional dependencies FD1 through FD4. (b) Decomposing into the 2NF relations LOTS1 and LOTS2. (c) Decomposing LOTS1 into the 3NF relations LOTS1A and LOTS1B. (d) Summary of the progressive normalization of LOTS.

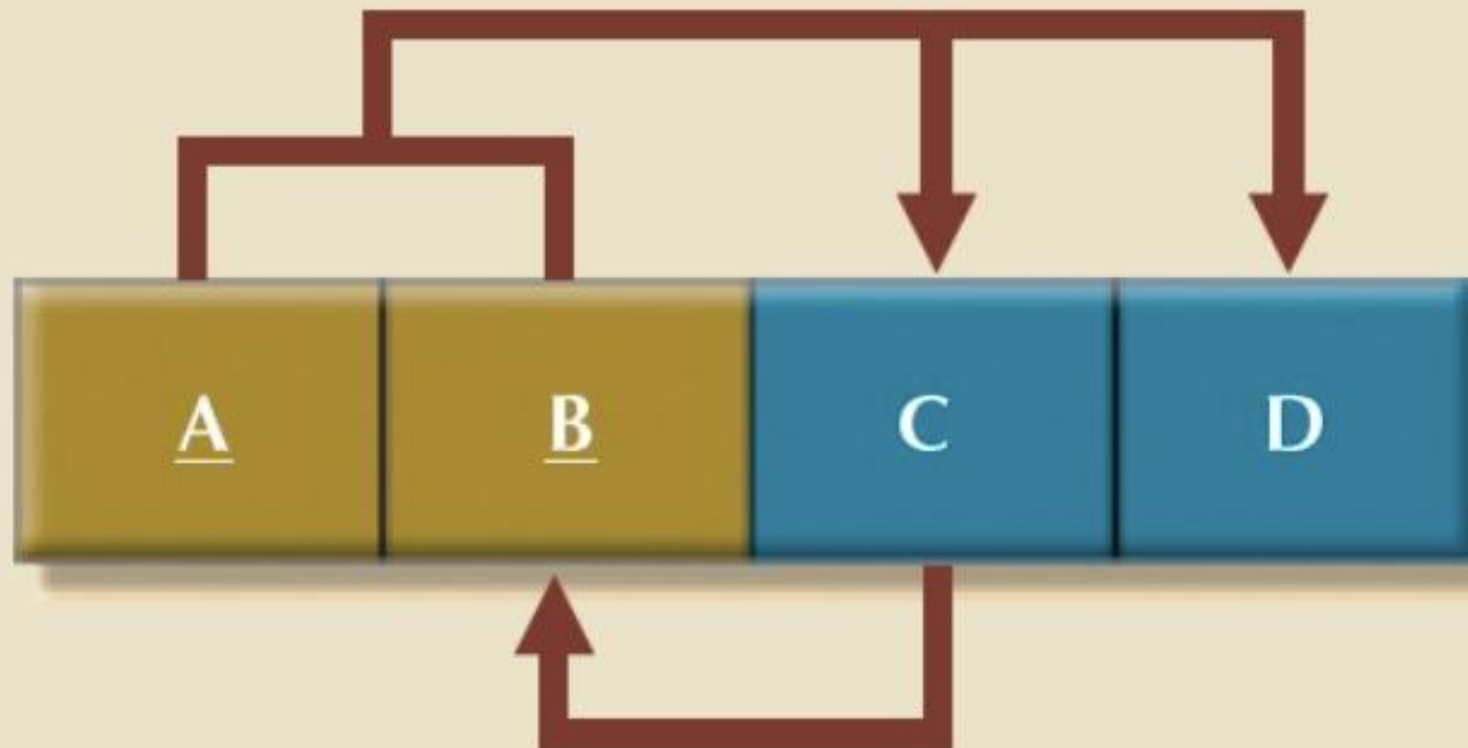
Boyce-Codd Normal Form

- A relation schema R is in Boyce-Codd Normal Form(BCNF) if whenever an FD $X \rightarrow A$ holds in R , then X is a superkey of R
- Each normal form is strictly stronger than the previous one
 - Every 2NF relation is in 1NF
 - Every 3NF relation is in 2NF
 - Every BCNF relation is in 3NF
- There exist relations that are in 3NF but not in BCNF
- Hence BCNF is considered a stronger form of 3NF
- The goal is to have each relation in BCNF (or 3NF)

Boyce-Codd Normal Form

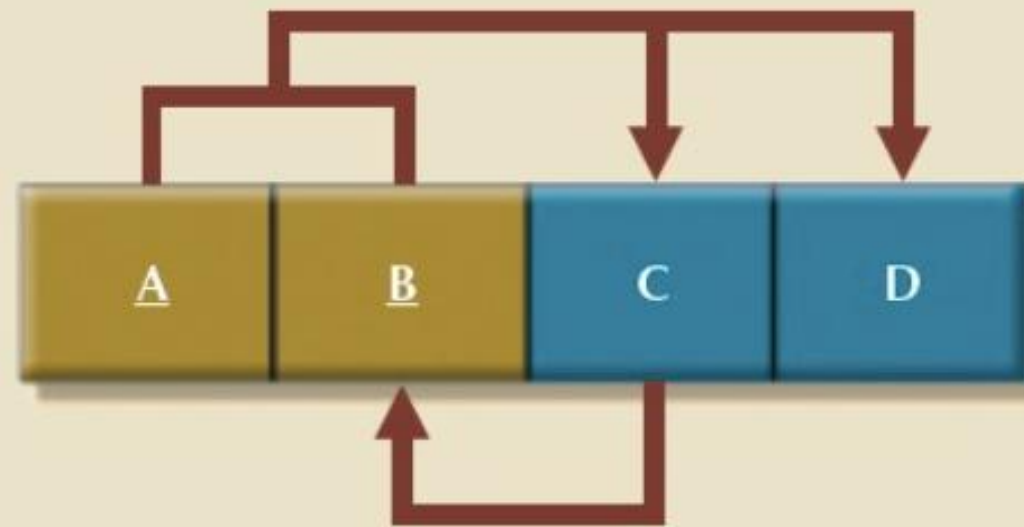
- Every determinant in the table should be a candidate key
 - Candidate key: same characteristics as primary key but not chosen to be the primary key
 - Equivalent to 3NF when the table contains only one candidate key
 - Violated only when the table contains more than one candidate key
 - Considered to be a special case of 3NF

A TABLE THAT IS IN 3NF BUT NOT IN BCNF

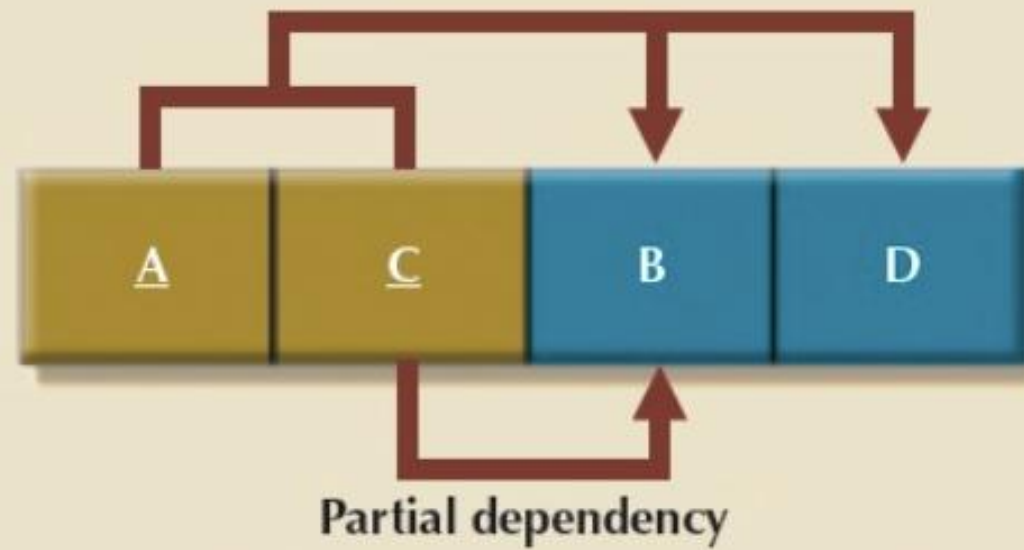


DECOMPOSITION TO BCNF

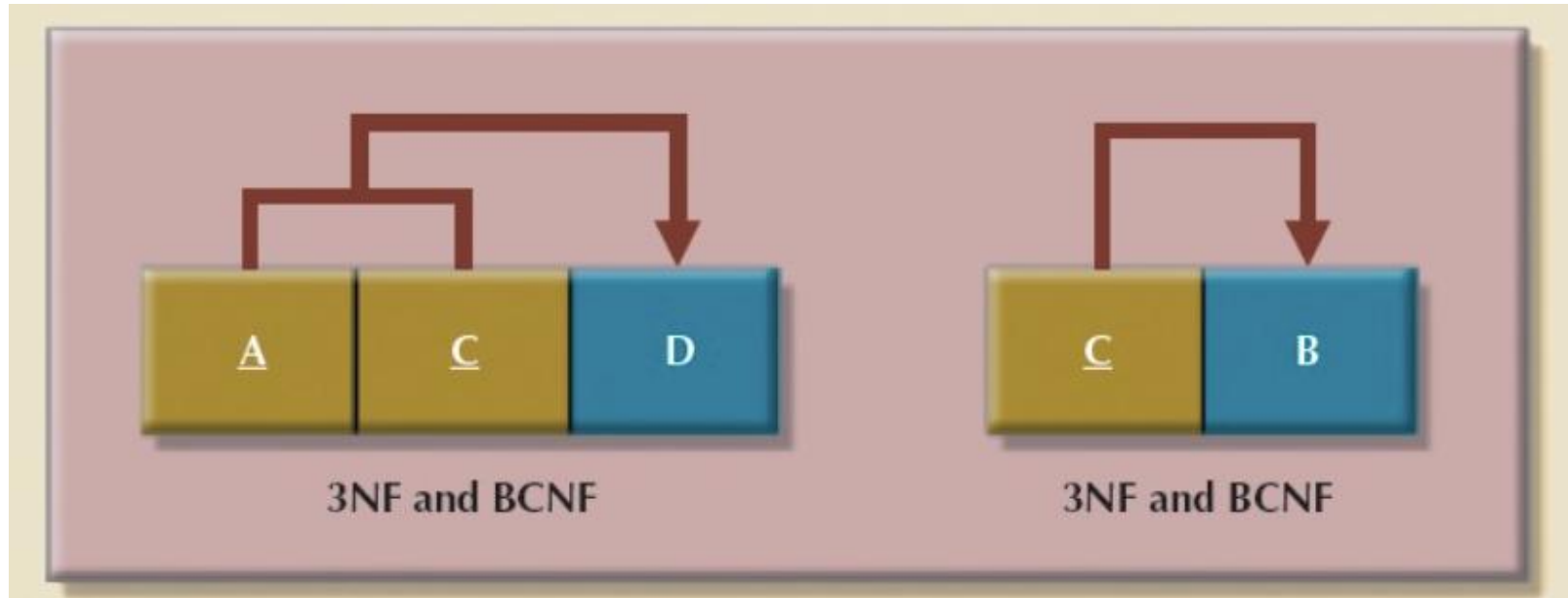
3NF, but not BCNF



1NF



AFTER DECOMPOSITION TO BCNF



Boyce-Codd normal form

Normalization into BCNF- example
from Ramez Elmasri and Shamkant
Navathe (2007)

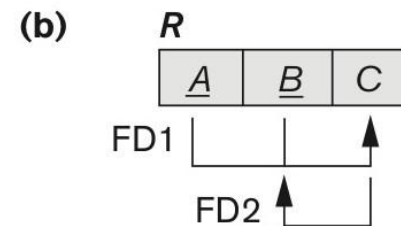
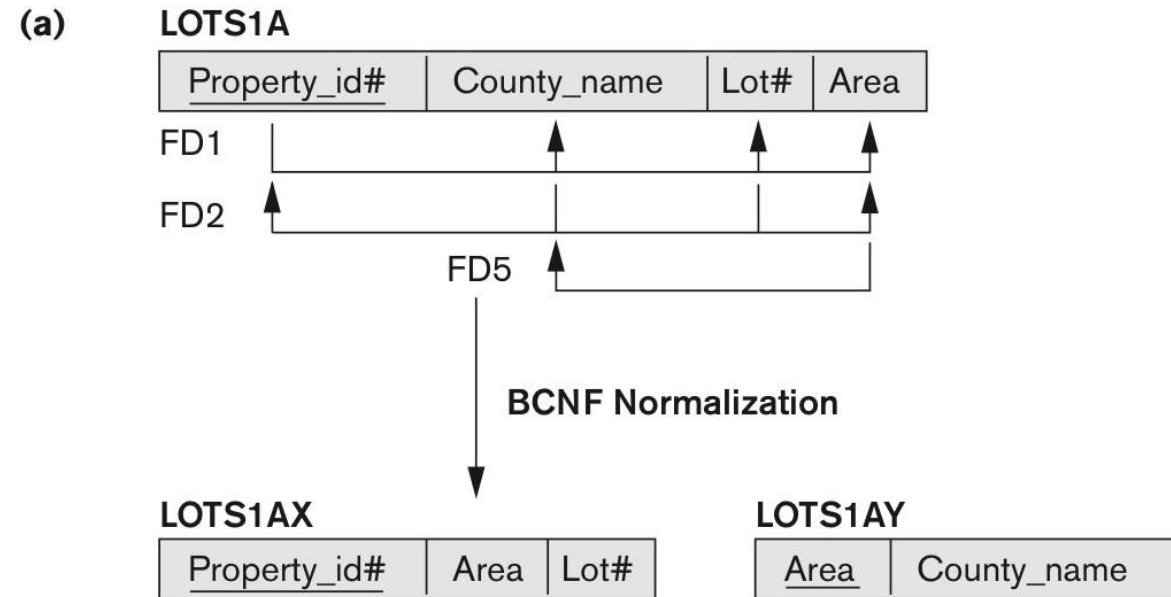


Figure 4.7
Boyce-Codd normal form. (a) BCNF normalization of LOTS1A with the functional dependency FD2 being lost in the decomposition. (b) A schematic relation with FDs; it is in 3NF, but not in BCNF due to the f.d. $C \rightarrow B$.

Figure 4.8 A relation TEACH that is in 3NF but not in BCNF

TEACH

Student	Course	Instructor
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe
Narayan	Operating Systems	Ammar

PK :fd1: { student, course} -> instructor
fd2: instructor -> course

Notice this pattern:

$A, B \rightarrow C$

$C \rightarrow B$

Figure 4.8

A relation TEACH that is in 3NF
but not BCNF.

Achieving the BCNF by Decomposition

Two FDs exist in the relation TEACH:

- fd1: { student, course } $\rightarrow$ instructor
- fd2: instructor $\rightarrow$ course

{student, course} is a candidate key for this relation and that the dependencies shown follow the pattern in Figure 4.7(b).

- So this relation is in 3NF *but not in* BCNF

A relation **NOT** in BCNF should be decomposed so as to meet this property, while possibly forgoing the preservation of all functional dependencies in the decomposed relations.

Fourth Normal Form (4NF)

A relation is in 4NF when it:

- Is in 3NF
- Has no multivalued dependencies

Rules

- All attributes must be dependent on the primary key, but they must be independent of each other
- No row may contain two or more multivalued facts about an entity

Multivalued Dependencies and Fourth Normal Form (1)

Definition:

A **multivalued dependency (MVD)** $X \twoheadrightarrow Y$ specified on relation schema R , where X and Y are both subsets of R , specifies the following constraint on any relation state r of R : If two tuples t_1 and t_2 exist in r such that $t_1[X] = t_2[X]$, then two tuples t_3 and t_4 should also exist in r with the following properties, where we use Z to denote $(R - (X \cup Y))$:

- $t_3[X] = t_4[X] = t_1[X] = t_2[X]$.
- $t_3[Y] = t_1[Y]$ and $t_4[Y] = t_2[Y]$.
- $t_3[Z] = t_2[Z]$ and $t_4[Z] = t_1[Z]$.

An MVD $X \twoheadrightarrow Y$ in R is called a **trivial MVD** if (a) Y is a subset of X , or (b) $X \cup Y = R$.

Multivalued Dependencies and Fourth Normal Form (3)

Definition:

A relation schema R is in **4NF** with respect to a set of dependencies F (that includes functional dependencies and multivalued dependencies) if, for every *nontrivial* multivalued dependency $X \twoheadrightarrow Y$ in F^+ , X is a superkey for R .

- Note: F^+ is the (complete) set of all dependencies (functional or multivalued) that will hold in every relation state r of R that satisfies F . It is also called the **closure** of F .

Figure 4.9 Fourth and Fifth normal forms.

(a) EMP

<u>Ename</u>	<u>Pname</u>	<u>Dname</u>
Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John

(b) EMP_PROJECTS

<u>Ename</u>	<u>Pname</u>
Smith	X
Smith	Y

EMP_DEPENDENTS

<u>Ename</u>	<u>Dname</u>
Smith	John
Smith	Anna

(c) SUPPLY

<u>Sname</u>	<u>Part_name</u>	<u>Proj_name</u>
Smith	Bolt	ProjX
Smith	Nut	ProjY
Adamsky	Bolt	ProjY
Walton	Nut	ProjZ
Adamsky	Nail	ProjX
Adamsky	Bolt	ProjX
Smith	Bolt	ProjY

Figure 4.9

Fourth and fifth normal forms. (a) The EMP relation with two MVDs: $Ename \twoheadrightarrow Pname$ and $Ename \twoheadrightarrow Dname$. (b) Decomposing the EMP relation into two 4NF relations EMP_PROJECTS and EMP_DEPENDENTS. (c) The relation SUPPLY with no MVDs is in 4NF but not in 5NF if it has the JD(R_1, R_2, R_3).

Summary of Normalization Theory

Source of Anomaly	Normal Form	Design Principles
Functional Dependencies (FDs)	1NF, 2NF, 3NF, BCNF	BCNF: Design tables so that every determinant is a candidate key
Multivalued dependencies (MVDs)	4NF	4NF: Move each multivalued dependency to a table of its own

Exercise: Perform Normalization until 3NF

DepartmentComplaint

Dept#	DeptName	Location	MgrName	MgrId	TelExt	Cust#	CustName	Dateof Complaint	Natureof Complaint
1123	Soap Divison	Ipoh	Uwais	M45	6543	C123	Izzah	24/4/09	Not enough bubbles
						C124	Uzair	14/6/09	Smell like fish

take a good look at the data and determine the primary key and other functional dependencies (FDs)

Dept#, Cust# → DeptName, Location, MgrName, MgrId, TelExt, CustName, DateOfComplaint, NatureOfComplaint

Dept# → DeptName, Location, MgrName, MgrId, TelExt

Cust# → CustName

MgrId → MgrName

Approaches to removing repeating groups

There are two approaches to removing repeating groups from unnormalized tables:

- Removes the repeating groups by entering appropriate data in the empty columns of rows containing the repeating data.
- Removes the repeating group by placing the repeating data, along with a copy of the original key attribute(s), in a separate relation. A primary key is identified for the new relation.

Split out repeating elements

DEPARTMENTCOMPLAINT

Dept#	DeptName	Location	MgrName	MgrId	TelExt	<u>Cust#</u>	CustName	Dateof Complaint	Natureof Complaint
1123	Soap Divison	Ipoh	Uwais	M45	6543	C123	Izzah	24/4/09	Not enough bubbles
1123	Soap Divison	Ipoh	Uwais	M45	6543	C124	Uzair	14/6/09	Smell like fish

DEPARTMENT

<u>Dept#</u>	DeptName	Location	MgrName	MgrId	TelExt
1123	Soap Divison	Ipoh	Uwais	M45	6543

COMPLAINT

<u>Dept#</u>	<u>Cust#</u>	CustName	Dateof Complaint	Natureof Complaint
1123	C123	Izzah	24/4/09	Not enough bubbles
1123	C124	Uzair	14/6/09	Smell like fish

2NF : Remove partial dependencies

DEPARTMENT

<u>Dept#</u>	DeptName	Location	MgrName	MgrId	TelExt
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COMPLAINT

<u>Dept#</u>	<u>Cust#</u>	CustName	Dateof Complaint	Natureof Complaint
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CUSTOMERCOMPLAINT

<u>Dept#</u>	<u>Cust#</u>	DateofComplaint	NatureofComplaint
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CUSTOMER

<u>Cust#</u>	CustName
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DEPARTMENT

<u>Dept#</u>	DeptName	Location	MgrName	MgrId	TelExt
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3NF : Remove transitive dependencies

CUSTOMERCOMPLAINT

<u>Dept#</u>	<u>Cust#</u>	DateofComplaint	NatureofComplaint
--------------	--------------	-----------------	-------------------

CUSTOMER

<u>Cust#</u>	CustName
--------------	----------

DEPARTMENT

<u>Dept#</u>	DeptName	Location	MgrName	MgrId	TelExt
--------------	----------	----------	---------	-------	--------

MANAGER

<u>MgrId</u>	MgrName
--------------	---------

CUSTOMER

<u>Cust#</u>	CustName
--------------	----------

DEPARTMENT

<u>Dept#</u>	DeptName	Location	MgrId	TelExt
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CUSTOMERCOMPLAINT

<u>Dept#</u>	<u>Cust#</u>	DateofComplaint	NatureofComplaint
--------------	--------------	-----------------	-------------------

Exercise— Perform Normalization until BCNF

Example of BCNF:

Suppose there is a college where one faculty teaches in more than one courses.

<u>F_id</u>	F_address	<u>Course_id</u>	Course_name
101	Putrajaya	C1	MCA
101	Putrajaya	C2	MBA
102	Kuala Lumpur	C1	MCA
102	Kuala Lumpur	C2	MBA

Exercise—Try transform till BCNF (cont)

Faculty_address

<u>F_id</u>	F_address
101	Putrajaya
102	Kuala Lumpur

Course

<u>Course_id</u>	Course_name
C1	MCA
C2	MBA

Exercise—Try transform till BCNF (cont)

Faculty_Course

<u>F_id</u>	<u>Course_id</u>
101	C1
101	C2
102	C1
102	C2

Exercise- Try to transform till 4NF

<u>student_id</u>	Name	<u>Course</u>
101	Ahmad	Python
102	Tini	Java
103	Zafrul	R programming
101	Ahmad	Java
105	Aizul	PHP

Exercise- Try to transform till 4NF (cont..)

Student_name	<u>student_id</u>	Name
	101	Ahmad
	102	Tini
	103	Zafrul
	105	Aizul

Student_course	<u>student_id</u>	<u>Course</u>
	101	Python
	102	Java
	103	R programming
	101	JAVA
	105	PHP

In a nutshell

A table is in first normal form if

there are no repeating groups

all the key attributes are defined

all attributes are dependent on the primary key

A table is in 2nd normal form if

it's in 1st normal form

it includes no partial dependencies (where an attribute is dependent on only a part of a primary key).
--

A table is in 3rd normal form if

It's in 2nd normal form

It contains no transitive dependencies (where a non-key attribute is dependent on another non-key attribute).

Chapter Summary

- Functional Dependencies (FDs)
- Normal Forms (1NF, 2NF, 3NF) Based on PrimaryKeys
- General Normal Form Definitions of 2NF and 3NF(For Multiple Keys)
- BCNF (Boyce-Codd Normal Form)
- Fourth Normal Form

Exercise 1

ClientNo	cName	propertyNo	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	John kay	PG4	6 lawrence St,Glasgow	1-Jul-00	31-Aug-01	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-02	1-Sep-02	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 lawrence St,Glasgow	1-Sep-99	10-Jun-00	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-00	1-Dec-01	370	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Nov-02	1-Aug-03	450	CO93	Tony Shaw

Exercise 2

<u>Student ID</u>	Student Name	<u>Major</u>	Advisor	Advisor City
123	Avi	Economics	Samuelson	Cambridge
123	Avi	Math	Nash	New Haven
456	Darin	Math	Nash	New Haven
789	Len	Physics	Einstein	Princeton
999	Richard	Physics	Bohr	Munich

Exercise 3

Sales Order

Fiction Company
202 N. Main
Manhattan, KS 66502

CustomerNumber: 1001
Customer Name: ABC Company
Customer Address: 100 Points
Manhattan, KS 66502

Sales Order Number: 405
Sales Order Date: 2/1/2000
Clerk Number: 210
Clerk Name: Martin Lawrence

Item Ordered	Description	Quantity	Unit Price	Total
800	widgit small	40	60.00	2,400.00
801	tingimajigger	20	20.00	400.00
805	thingibob	10	100.00	1,000.00
Order Total				3,800.00

Fields in the original data table will be as follows:

SalesOrderNo, Date, CustomerNo, CustomerName, CustomerAdd, ClerkNo,
ClerkName, ItemNo, Description, Qty, UnitPrice

Decompose the relation until to **3NF**.